

Population and Economic Growth When Malthus Meets Romer

Anson Zhou

The University of Hong Kong

October 2026

One of the Oldest Debates in Economics

- Does a larger population raise or lower living standards?
 - Malthus (1798): population strains resources $\Rightarrow$ wages to subsistence
 - * David Ricardo, Paul Ehrlich, William Catton, . . .
 - Romer (1990): more people $\Rightarrow$ more ideas $\Rightarrow$ innovation dominates
 - * Richard Jones, Julian Simon, Chad Jones, Elon Musk, . . .
- Never more consequential: 2bn to 8bn+ in a century, now fertility below replacement across the developed world
 - “The end of economic growth” – Jones (2022)
- This paper: study the implications of the [disagreement itself](#)

This Paper

1. OLG Model: Barro–Becker altruism + semi-endogenous growth + heterogeneous beliefs about population scale effects
2. Mechanism: beliefs shape fertility $\Rightarrow$ fertility shapes evidence $\Rightarrow$ wrong beliefs can persist and delay growth
3. Theoretical predictions: fertility ranking between types is state-dependent; wrong beliefs generate cycles and traps
4. Evidence: Malthusian beliefs correlated with ≈ 0.3 fewer children (GSS) + Malthus at Haileybury (Robertson 2026)
5. Calibration: inherited disagreement delays takeoff by $\sim 1,125$ years; information trap if initial Romerian share $\leq 8\%$

Related Literature & Contribution

- **Fertility & unified growth:** Barro–Becker (1989), Galor (2000, 2011), Hansen–Prescott (2002); demographic transition and sustained growth
- **Scale effects:** Romer (1990), Kremer (1993), Jones (1995, 2022); the sign of the population–technology link is central to long-run projections
- **Heterogeneous beliefs:** Angeletos–La’O (2013), Coibion–Gorodnichenko (2015); climate disagreement (Bakkensen–Barrage 2022)
- **Cultural transmission, cultural entrepreneur & evolutionary selection:** Bisin–Verdier (2001, 2011), Galor–Moav (2002), Galor–Michalopoulos (2012), Mokyr (2016)

This paper: disagreement about the [sign](#) of the population scale effect, where fertility choices shape both the demographic composition of beliefs and the evidence available to evaluate them

Roadmap

- Model
- Empirical evidence: beliefs and fertility
- Equilibrium dynamics
- Application with quantitative illustration

Model Setup

Model: Environment

- Discrete-time OLG; continuum of atomistic households
- Time endowment 1; fertility n costs time $\psi(n)$, labor supply $1 - \psi(n)$
- Linear production: $Y_t = A_t L_t$, wage = A_t , so

$$c_{j,t} = A_t(1 - \psi(n_{j,t}))$$

- Population and its adjusting reference:

$$N_{t+1} = \bar{n}_t N_t, \quad \bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}$$

- ϕ_t = share of Romerians at time t

- Productivity growth:

$$\ln A_{t+1} - \ln A_t = \theta^* x_t + \eta_{t+1}, \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma^2)$$

- Population gap x_t : population relative to an adjusting reference, evolves as

$$x_{t+1} = (1 - \mu)x_t + \ln \bar{n}_t, \quad \mu \in (0, 1]$$

- Disagreement: $\theta^* \in \{\theta_R, \theta_M\}$, $\theta_R > 0 > \theta_M$
- μ = catch-up rate of the reference (semi-endogenous, Jones 1995):
 - one-time population increase $\Rightarrow$ temporary growth effect
 - sustained growth $\bar{n} > 1 \Rightarrow$ permanent: $g_\infty = \theta^* \ln \bar{n} / \mu$
- Disagreement is about the **sign** of θ^*

Key Structural Property: The Uninformative Reference

- At $x = 0$: both models predict $\mathbb{E}[\Delta \ln A] = 0$ — same mean, same variance
- Signal informativeness (KL divergence):

$$\mathcal{I}_t = \frac{(\theta_R - \theta_M)^2 x_t^2}{2\sigma^2} \implies \mathcal{I}_t = 0 \text{ at } x_t = 0$$

- When $x_t = 0$, observed growth data **cannot distinguish** between worldviews
- Information increases with the squared distance from the reference — gap half as large $\Rightarrow$ one quarter the information

- Dynastic utility:

$$V_{jt} = \max_{n_{jt}} u(c_{jt}) + \beta n_{jt}^{1-\varepsilon} \mathbb{E}_j[V_{j',t+1}], \quad \rho, \varepsilon \in (0, 1)$$

subject to $c_{jt} = A_t(1 - \psi(n_{jt}))$

- Time cost of childbearing: increasing and convex
- CRRA utility + time costs $\Rightarrow$ TFP level cancels from the fertility FOC
- Beliefs matter only via expected growth — where the disagreement lives

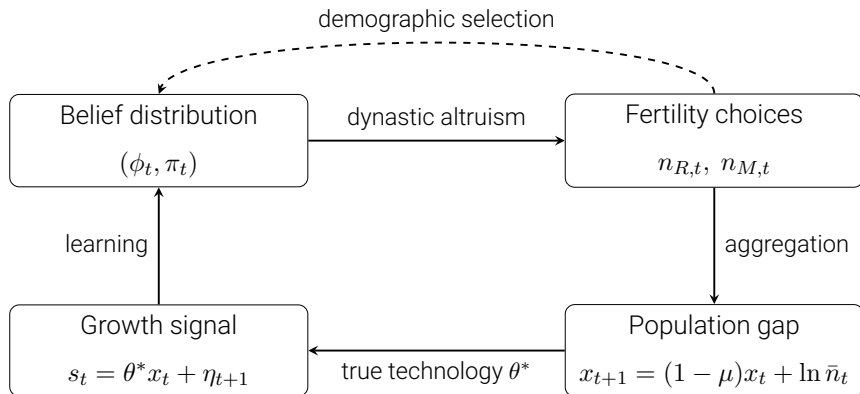
Beliefs, Learning, and Transmission

- Two types: share ϕ_t Romerians ($\theta_R > 0$), $1 - \phi_t$ Malthusians ($\theta_M < 0$)
- Each parent chooses fertility using their own worldview
- Social credence π_t : shared posterior, Bayes-updated from growth signals
 $s_t = \theta^* x_t + \eta_{t+1}$
- Transmission: child inherits parent's worldview with prob λ ; otherwise adopts the Romerian view with prob π_{t+1}
- Belief distribution evolves via two channels:

$$\phi_{t+1} = \underbrace{\lambda \phi_t^{birth}}_{\text{cultural transmission}} + \underbrace{(1 - \lambda) \pi_{t+1}}_{\text{learning-based recruitment}}$$

- Reproductive selection and social learning can conflict

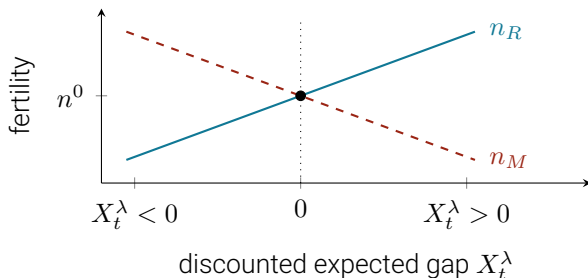
The Belief–Fertility–Growth Feedback Loop



- Near $x = 0$: both feedback channels into beliefs shut down together
- Household choices shape the evidence that future generations use to revise beliefs

State-Dependent Fertility Ranking

$$n_{R,t} - n_{M,t} \approx \underbrace{\frac{1-\rho}{1+\varepsilon}}_K \cdot \Delta\theta \cdot X_t^\lambda \quad \text{where} \quad X_t^\lambda \equiv \mathbb{E}_t \left[\sum_{k \geq 0} (\delta\lambda)^k x_{t+k} \right]$$



- $X_t^\lambda > 0$ (above reference on balance): Romerians out-reproduce
- $X_t^\lambda < 0$ (below reference on balance): ranking flips
- At $X_t^\lambda = 0$: identical fertility despite disagreement

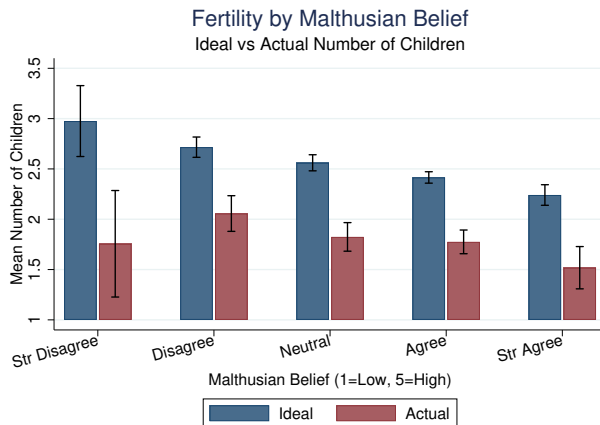
Empirical Support

Beliefs and Fertility: GSS Evidence

proxy

regressions

- GSS 2000 & 2010 ($N = 1,822$): agree/disagree “Earth cannot continue to support population growth” – 55% Malthusian
- Malthusians desire 0.27 fewer children (ideal) and have 0.29 fewer (actual)
- Gradient present across the full belief intensity scale



Regression Evidence

by education

completed fertility

	Ideal Fertility		Actual Fertility	
	Baseline	Full	Baseline	Full
Malthusian	−0.268*** (0.041)	−0.203*** (0.045)	−0.292*** (0.066)	−0.237*** (0.071)
Demographic controls + Year FE	Yes	Yes	Yes	Yes
Additional controls	No	Yes	No	Yes
Observations	1,822	1,567	1,822	1,567
R^2	0.060	0.099	0.262	0.333

- Baseline: age, age², sex, education, year FE
- Full adds: marital status, race, religion, political views, income
- Full-controls estimates preserve $\approx 75\%$ of baseline association

Extensive Margin and Reverse Causality

[details](#)[age groups](#)

Margin decomposition:

- Extensive: Malthusian odds ratio = 0.55 (AME: -9.2 pp, $p < 0.001$)
- Intensive (conditional on > 0): -0.12 ($p = 0.089$)
- Association concentrated on the decision to become a parent

Reverse causality checks:

- Ideal-fertility gradient significant in all age groups, including 18–34 (limited childbearing experience)
- Actual-fertility: zero among young, large among 35+
- Siblings control: coefficient changes $< 5\%$
- Environmental concern: Malthusian coefficient unchanged

- Pattern consistent with beliefs $\rightarrow$ fertility, not the reverse

Complementary Evidence: Malthus at Haileybury

- Robertson (2026): British Indian officials trained at Haileybury, where Malthus himself taught political economy
- Malthus's abrupt death $\Rightarrow$ replaced by Richard Jones (a critic) $\Rightarrow$ quasi-exogenous variation in doctrine across cohorts
- Officials taught by Malthus provided systematically less famine relief, decades later
- Three features: exogenous assignment, costly behavior (not cheap talk), decades-long durability
- Complements GSS: clean identification of the belief channel, though in bureaucratic relief rather than fertility

Reading the Evidence Through the Model

- Model prediction is conditional: $n_R > n_M$ iff $X_t^\lambda > 0$
- In 2000/2010: world population salient and growing $\Rightarrow X_t^\lambda > 0$ plausibly satisfied $\Rightarrow$ negative gradient is exactly this case
- Magnitudes: all-Romerian vs. all-Malthusian gap ≈ 0.29 children
 $\Rightarrow$ 10 pp composition shift ≈ 0.03 fewer children
- Out-of-sample prediction: gradient should **reverse** where depopulation is the salient scenario (e.g., East Asia) – but which scale is relevant?

Equilibrium Dynamics

Demographic Selection and Learning

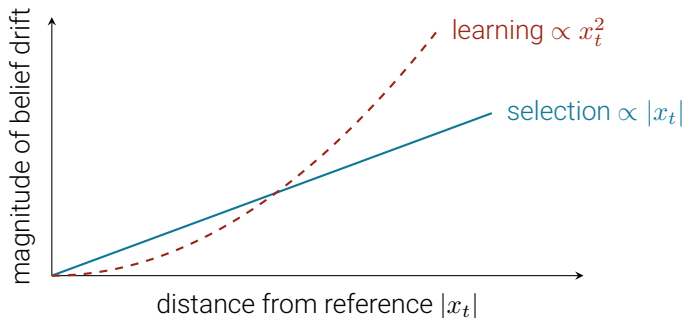
- Expected change in adult Romerian share:

$$\mathbb{E}_t[\phi_{t+1} - \phi_t] = \underbrace{\lambda \frac{\phi_t(1 - \phi_t)(n_{R,t} - n_{M,t})}{\bar{n}_t}}_{\text{reproductive selection}} + \underbrace{(1 - \lambda)(\pi_t - \phi_t)}_{\text{recruitment to credence}} + \underbrace{(1 - \lambda)\mathbb{E}_t[\pi_{t+1} - \pi_t]}_{\text{new evidence}}$$

- Selection favors trajectory-optimists; learning favors the truth

Expected path	Malthusian world	Romerian world
Above reference	Selection: Romerians ✗	Selection: Romerians ✓
	Learning: Malthusians ✓	Learning: Romerians ✓
Below reference	Selection: Malthusians ✓	Selection: Malthusians ✗
	Learning: Malthusians ✓	Learning: Romerians ✓

Two Engines of Belief Change



- Selection favors trajectory-optimists — vanishes linearly with $|x_t|$
- Learning favors the truth — vanishes quadratically
- Near reference: selection dominates; far away: learning dominates
- At $x = 0$: both exactly zero — disagreement is self-sustaining

Information Accumulation

- Cumulative signal information: $\mathcal{C}_T = \sum_{t=0}^{T-1} \mathcal{I}_t \propto \sum_{t=0}^{T-1} x_t^2$
- Theorem (Information and posterior convergence):
 - On $\{\sum_t \mathcal{I}_t = \infty\}$: social credence $\pi_t \rightarrow$ truth a.s.
 - On $\{\sum_t \mathcal{I}_t < \infty\}$: $\pi_t \rightarrow$ interior limit $\pi_\infty \in (0, 1)$
- Sustained fertility above or below replacement $\Rightarrow$ gap stays away from zero $\Rightarrow$ infinite info $\Rightarrow$ truth identified
- Geometric return to reference $\Rightarrow \sum x_t^2 < \infty \Rightarrow$ finite info $\Rightarrow$ persistent disagreement
- Whether learning resolves the debate depends on the population path

The Dormant Informational Trap

formal statement

- Theorem (Dormancy):

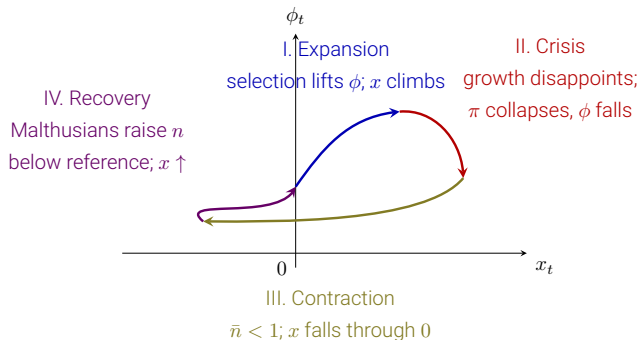
- Exact: $x_0 = 0 \Rightarrow x_t = 0, n_{R,t} = n_{M,t} = 1, \pi_t = \pi_0$ forever; wrong beliefs never corrected
- Near-dormant: for $|x_0|$ small, with high probability:

$$|x_t| \leq q^t |x_0|, \quad \sum_t \mathcal{I}_t < \infty, \quad \pi_t \rightarrow \pi_\infty \in (0, 1)$$

- Population converges to a finite level while leaving the scale-effect disagreement unresolved
- Key asymmetry:
 - Optimism exposes itself (sustained growth keeps data informative)
 - Pessimism **hides the evidence** that would correct it

Malthusian World: Boom-Bust Cycles

cycle mechanics



- Selection and learning oppose above the reference $\Rightarrow$ tug-of-war
- Busts reinforced by selection; recoveries engineered by pessimists (they raise fertility when $x < 0$)
- Damped cycles: $\sum x_t^2 < \infty$ possible $\Rightarrow$ dormancy with accumulated belief

Romerian World: Dormancy and Escape

trap regimes

- Above reference on balance: selection and learning aligned — both favor Romerians $\Rightarrow$ self-cementing growth
- Below reference on balance: Malthusians gain through reproduction while evidence favors Romerians — two forces conflict
- Near the reference: both forces vanish $\Rightarrow$ **dormant trap**
 - $\rightarrow$ Fertility slightly above replacement $\Rightarrow$ composition slowly shifts toward Romerians $\Rightarrow$ trap eventually destabilizes
 - $\rightarrow$ Fertility slightly below replacement $\Rightarrow$ deeper pessimism $\Rightarrow$ signals get quieter $\Rightarrow$ deepening spiral
- Once escaped: sustained growth keeps data informative, truth locks in

Quantitative Findings

A Correct-Belief Benchmark

- Compare two economies with the same true technology, initial gap $x_0 = z > 0$, and shocks:
 - Economy C : all households know the truth ($\phi = \pi = 1$)
 - Economy H : inherited disagreement ($\phi_0 = \pi_0 = p < 1$)
- For a threshold $h > z$, define first entry time $T_h^k = \inf\{t : x_t^k \geq h\}$
- **Proposition (Entry delay)**: Suppose truth is Romerian.
 - Economy C : correct beliefs amplify the displacement $\Rightarrow T_h^C < \infty$
 - Economy H : inherited beliefs contract the gap back toward zero $\Rightarrow \mathbb{P}(T_h^H = \infty) > 0$ & the probability is decreasing in z
- The same displacement produces **opposite responses**: amplification vs. contraction — entirely attributable to inherited beliefs

Application: Delayed Takeoff

- Puzzle: why the long delay between available knowledge and economic takeoff?
- Model: at $x \approx 0$, a Romerian world looks exactly like a Malthusian one
 - Land-constrained history supports Malthusian beliefs (Clark 2007)
 - After regime change: near reference, rival models predict the same growth
- Two escape channels:
 1. Compositional selection: if $n^0 > 1$, small $x^* > 0 \Rightarrow$ selection slowly favors Romerians $\Rightarrow \phi$ drifts up until trap destabilizes
 2. Cultural shift (Mokyr 2016): exogenous surge of optimism, not from macro data but from ideas
- Once escaped, self-cementing: sustained growth keeps data informative

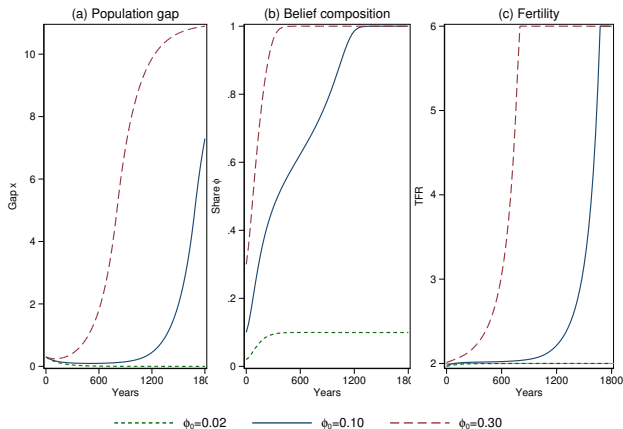
Calibration

computation details

Parameter	Value	Target / source
ρ, ε	0.50	CRRA & dynastic curvature (Jones–Schoonbroodt)
γ	0.312	15% child-rearing time at replacement
β	0.252	Replacement normalization $\Rightarrow \delta \simeq 0.253$
σ	0.20	$\approx 4\%$ annual s.d. of productivity
μ	0.10	Gap half-life ≈ 165 years (Jones 1995, 2022)
θ_R	0.50	Multiplier $\theta_R/\mu = 5$ (Kremer 1993); 2%/yr at entry
θ_M	-0.15	Perceived elasticity $-\theta_M/\mu = 1.5$ (Lee 1980, Clark 2007)
λ	0.60	Vertical transmission (Bisin–Verdier 2001, 2011)
x_0	0.30	Initial displacement

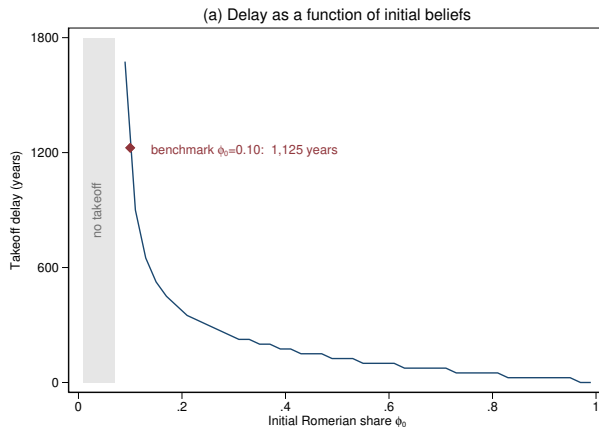
Threshold Dynamics

Monte Carlo



- $\phi_0 = 0.30$: fast takeoff (generation 20, ~ 500 years)
- $\phi_0 = 0.10$: gap contracts to 0.09, composition crosses p^* at gen. 20; takeoff at gen. 55 ($\sim 1,375$ yrs)
- $\phi_0 = 0.02$: trapped – credence converges to < 1 , gap returns to zero

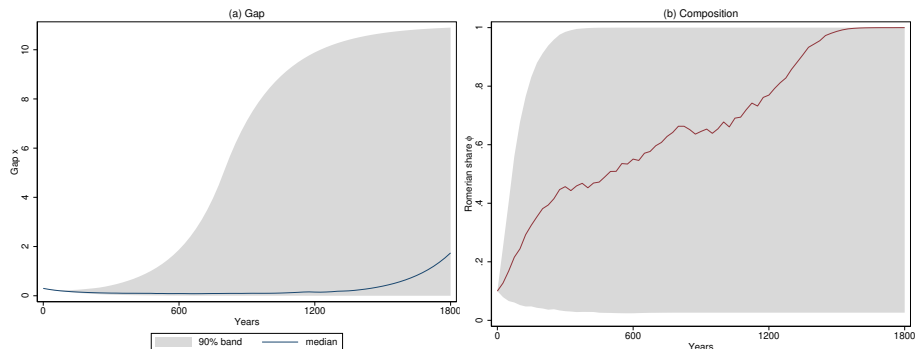
Takeoff Delay and the Cost of Disagreement



- $\phi_0 \leq 0.07$: no takeoff within 2,000 years (numerical dormant trap)
- Delay falls steeply: 1,675 yrs at $\phi_0 = 0.09$; < 300 yrs at $\phi_0 = 0.25$
- Welfare cost: $\omega \simeq 0.74\%$ of consumption per generation (20-gen window)

Monte Carlo: The Role of Luck

sensitivity



- 500 replications, benchmark $\phi_0 = 0.10$; takeoff timing widely dispersed
- Fastest 5%: entry < 500 years; bottom 5%: trapped forever
- The trap arises with positive probability even under the Romerian truth

Conclusion

- Disagreement about the sign of population scale effects generates:
 - state-dependent fertility rankings (validated in GSS)
 - boom-bust cycles in Malthusian worlds
 - an absorbing **dormant informational trap** in Romerian worlds
- Pessimism delays takeoff: inherited beliefs contract the gap and starve future generations of evidence
 - Calibrated delay: $\sim 1,125$ years; dormant trap below $\phi_0 \approx 0.08$
- Generalizable mechanism: optimal responses to beliefs generate endogenous information acquisition as an equilibrium outcome

Ecological vs. Economic Malthusianism

[back](#)

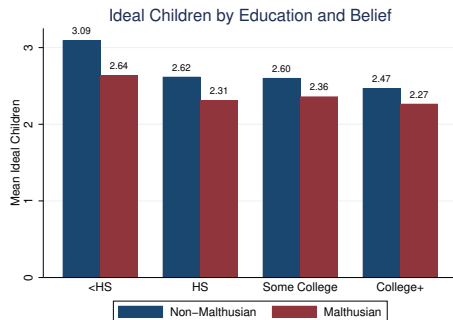
- GSS question is in environment modules: captures ecological concern (resource depletion, sustainability)
- For the fertility margin, distinction is second-order: A_t is broad TFP, and both idioms point to $\theta < 0$, a poorer future for descendants
- Including a measure of general environmental concern leaves the Malthusian coefficient essentially unchanged
- For historical claims the distinction is more substantive; the survey only validates the belief-to-fertility arrow

Regression Evidence: Continuous Scale and Counts

[back](#)

	Ideal Fertility	Actual Fertility
Malthusian (binary)	−0.268*** (0.041)	−0.292*** (0.066)
Malthusian (1–5 scale)	−0.153*** (0.021)	−0.146*** (0.033)
Poisson (binary)	−0.107*** (0.016)	−0.179*** (0.037)
Excl. neutrals (binary)	−0.357*** (0.054)	−0.342*** (0.087)
Controls + Year FE	Yes	Yes
Observations	1,822	1,822

Gradient Holds Within Education Groups

[back](#)

	Ideal		Actual	
	HS or Less	College+	HS or Less	College+
Malthusian	-0.318*** (0.069)	-0.235*** (0.049)	-0.209* (0.115)	-0.392*** (0.081)

Completed Fertility

[back](#)

	Full Sample	Completed	Not Yet Completed
Malthusian	−0.292*** (0.066)	−0.257** (0.109)	−0.288*** (0.079)
Observations	1,822	848	974

- Completed = women 45+, men 50+; all controls included
- Estimate stable: not driven by incomplete childbearing among the young

Extensive vs. Intensive Margin

[back](#)

	Extensive: Any Children		Intensive: Children > 0	
	Baseline	Full	Baseline	Full
Logit (OR)	0.554*** (0.067)	0.533*** (0.080)		
AME	-0.092*** (0.019)	-0.080*** (0.019)		
OLS			-0.122* (0.072)	-0.085 (0.080)
<i>N</i>	1,822	1,567	1,299	1,120

Association concentrated on the decision to become a parent.

Reverse Causality Checks

[back](#)

	Full Sample	Age 18–34	Age 35–49	Age 50+
<i>Panel A: Ideal Fertility</i>				
Malthusian	−0.268*** (0.041)	−0.243*** (0.075)	−0.322*** (0.079)	−0.253*** (0.061)
<i>Panel B: Actual Fertility</i>				
Malthusian	−0.292*** (0.066)	−0.052 (0.094)	−0.469*** (0.124)	−0.263** (0.114)
<i>N</i>	1,822	514	541	767

- Ideal gradient present even among young adults with little childbearing experience
- Actual: zero among 18–34, large among 35+ → consistent with beliefs affecting fertility

Why Each Preference Ingredient Is Load-Bearing

[back](#)

- Time cost + CRRA: TFP level cancels; income-supports-children channel off by construction — beliefs enter only via growth forecasts
- $\rho < 1$: required for positive dynastic value; $\rho > 1$ (with $\varepsilon > 1$) flips gradient sign — rejected by data
- $\varepsilon \in (0, 1)$: total weight on children increases with n at diminishing rate; well-posed fertility problem
- Imperfect transmission: $\delta\lambda$ discounting in the fertility differential; association between parent's type and distant descendants decays
- Replacement normalization: $\beta^\dagger$ set so $n^0 = 1$; implied $\delta \simeq 0.253$

Microfoundation of the Population Gap

[back](#)

- Jones idea production: $\ln(A_{t+1}/A_t) = \nu + \lambda \ln N_t - \omega \ln A_t + \eta_{t+1}$; defining x_t appropriately yields $\mu = \omega\theta$
- Limit $\mu \rightarrow 0$ reproduces Kremer–Romer strong-scale-effects; trap reverts to slow learning
- Absorbing trap is a genuine semi-endogenous phenomenon
- θ encodes growth channel net of congestion; level channel neutralized by preferences
- The reference index H_t adjusts gradually: $\ln H_{t+1} - \ln H_t = \mu x_t$
- Convergence of the gap to zero can occur at a population level different from the initial value

Local Equilibrium Response: Details

[back](#)

- The local gap coefficient solves:

$$(a(p) - (1 - \mu))(1 - \delta a(p)) = K\bar{\theta}_p$$

- Explicit root: $a(p) = \frac{1 + \delta(1 - \mu) - \sqrt{[1 - \delta(1 - \mu)]^2 - 4\delta K\bar{\theta}_p}}{2\delta}$
- Individual policy slopes: $\left. \frac{\partial n_j}{\partial x} \right|_0 = b(p) + \frac{K(\theta_j - \bar{\theta}_p)}{1 - \delta\lambda a(p)}$
- Amplification threshold: $a(p) = 1$ when $\bar{\theta}_p = \bar{\theta}^{dest} = \frac{\mu(1 - \delta)}{K}$
- At baseline: $\bar{\theta}^{dest} \simeq 0.224$, requiring Romerian share $p^* \simeq 0.58$

Dormant Trap: Formal Statement

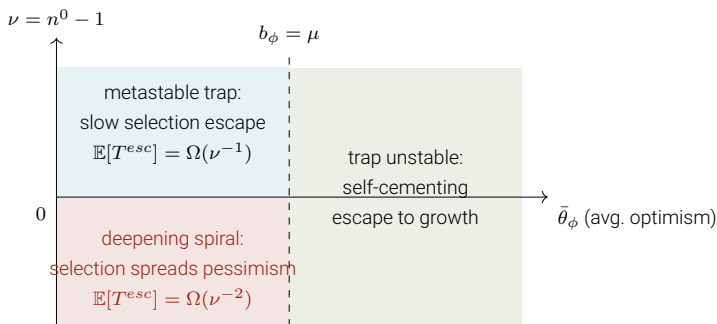
[back](#)

- Require uniform local return: $|F_x(0, \phi, \pi)| \leq q_0 < 1$ throughout a belief neighborhood
- On no-exit event E (probability $\geq 1 - C|x_0|/d$):

$$|x_t| \leq q^t |x_0|, \quad \sum_{t=0}^{\infty} \mathcal{I}_t \leq \frac{\Delta \theta^2 x_0^2}{2\sigma^2(1 - q^2)}$$

- Limiting credence is close to initial: $\mathbb{E}[\mathbb{1}_E |\pi_\infty - p|] \leq C|x_0|$
- With $\lambda < 1$: adult composition converges to π_∞
- Population converges to a positive finite level
- Long-run average log-productivity growth equals zero a.s. on E

Trap Regimes Diagram

[back](#)

- $\nu > 0$: selection slowly favors Romerians; eventual escape
- $\nu < 0$: selection favors Malthusians; escape time quadratic in $|\nu|$

Damped Cycles and Periodic Orbits

[back](#)

- Corollary (Damped demographic cycles): under uniformly negative $a(p) \in (-1, 0)$:

$$\text{sgn}(x_{t+1}) = -\text{sgn}(x_t), \quad |x_t| \leq q^t |x_0|$$

Population growth alternates; cumulative info finite; posterior interior

- Proposition (No periodic orbits under mean signals): replacing each signal by $s_t = \theta^* x_t$, no nonconstant periodic orbit with interior credence exists
 - Every nonzero gap adds evidence for the truth
 - Recurrent fixed-amplitude cycles require forgetting or outside shocks
- Damped-overshoot condition: $-\frac{(2-\mu)(1+\delta)}{K} < \bar{\theta}_p < -\frac{1-\mu}{K}$

- Fertility implements first-order log-form response:

$$\ln n_{j,t} = \nu + K [\bar{\theta}_p X_t^1 + (\theta_j - \bar{\theta}_p) X_t^\lambda]$$

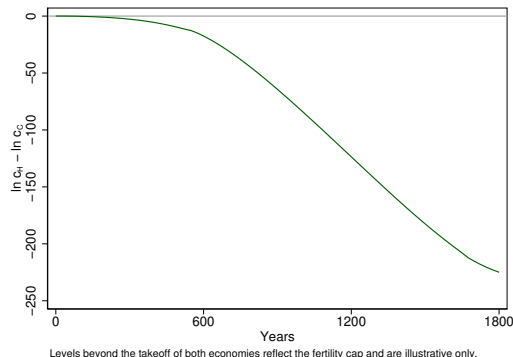
- X_t^λ, X_t^1 evaluated via constant-fertility forecast rule
- Aggregate fertility: within-period fixed point (local slope ≈ 0.07 , tolerance 10^{-12})
- TFR range: $[0.6, 6]$; beliefs capped at $(10^{-9}, 1 - 10^{-9})$
- Modern θ_R remark: semi-endogenous estimates (Jones 2022, Bloom et al. 2020) imply $\theta \approx 0.03 \ll \bar{\theta}^{dest}$ — even correct beliefs would not amplify
- Historical calibration uses strong-scale regime of Kremer (1993)

Welfare Construction and Consumption Paths

[back](#)

Welfare measure:

- Dynastic consumption-equivalent over 20-generation window
- $\omega = (V_0^C / V_0^H)^{1/(1-\rho)} - 1$
- Benchmark: $\omega \simeq 0.74\%$ per generation
- Stable across $\lambda \times \sigma$ grid (0.71–0.75%)
- Conservative: $\delta \simeq 0.25$ per gen discounts distant future



Log-consumption difference $H - C$: positive early (fewer children $\Rightarrow$ more time), negative and widening thereafter.

Parameter Sensitivity

[back](#)

Parameter variation	Delay (years)	Mechanism
$\lambda = 0.4$	950	Faster recruitment to credence
$\lambda = 0.6$ (baseline)	1,125	—
$\lambda = 0.8$	>1,775	Family transmission insulates
$\sigma = 0.1$	175	4× more info per observation
$\sigma = 0.2$ (baseline)	1,125	—
$\sigma = 0.3$	>1,775	Too noisy to learn
$\theta_M = -0.075$	975	Weaker contraction but less info
$\theta_M = -0.15$ (baseline)	1,125	—
$\theta_M = -0.30$	1,225	Stronger contraction offsets more info

- Transmission and noise interact strongly: high λ + high $\sigma \Rightarrow$ censored
- θ_M moves delay modestly: contraction and info effects largely offset

Robustness: What Would Break the Results

[back](#)

- No common crossing: rival models always distinguishable $\Rightarrow$ learning always active; trap vanishes
- $\rho > 1$: all rankings reverse, gradient flips — rejected by data
- $\mu \rightarrow 0$: trap reverts to slow learning; absorbing character requires $\mu > 0$
- Vanishing noise: trap dissolves if $\sigma^2(x)$ shrinks as fast as the squared gap
- Outside evidence: regional comparisons or research can identify θ^* without domestic demographic departure
- Persistent demographic shocks: can repeatedly displace economy into informative states, restoring learning
- Type-specific preferences: would preserve fertility differences at the reference $\Rightarrow$ exact dormancy breaks